SEVOFLURANE

Malignant hyperthermia trigger

Smooth inhalational induction

CHEMICAL NAME 1,1,1,3,3,3-hexafluoro-2-(fluoromethoxy)propane
CLASS Volatile inhalational general anaesthetic (halogenated ether)
TRADE NAMES Sevorane · Ultane · Sojourn

THERAPEUTIC USES / INDICATIONS

- Induction (especially children — pleasant smell, non-irritant) and maintenance of general anaesthesia.

ADMINISTRATION

- **Inhalation** via calibrated agent-specific vaporiser in anaesthetic machine.
- Concentration titrated by MAC (minimum alveolar concentration).
- Scavenging system required to protect staff.

DOSAGES

- **Induction:** up to 8% in oxygen ± nitrous oxide.
- **Maintenance:** 0.5–3% (MAC ~2% in adults; higher in infants, lower in elderly).
- Use fresh-gas flow ≥1–2 L/min to limit compound A formation.

PHARMACOKINETICS

- **Onset:** 1–2 min (low blood solubility). **Recovery:** rapid, 5–10 min after stopping.
- ~3–5% metabolised in liver (CYP2E1) → fluoride ions.
- Eliminated mainly unchanged via the lungs.

PHARMACODYNAMICS

- Enhances inhibitory **GABA-A** and glycine receptors; inhibits excitatory channels → unconsciousness, amnesia, immobility.
- Dose-dependent ↓ BP and respiratory drive; bronchodilator.
- Potentiates neuromuscular blockers.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea and vomiting after anaesthesia.
- Hypotension, bradycardia.
- Shivering; cough on emergence.

ADVERSE EFFECTS (SERIOUS)

- Emergence delirium/agitation in children.
- Malignant hyperthermia (rare, genetic): ↑ $ETCO_2$, rigidity, hyperthermia → treat with dantrolene.
- QT prolongation.
- Hepatotoxicity (very rare); nephrotoxicity concern with low flows.

CONTRAINDICATIONS & CAUTIONS

- Known or suspected susceptibility to malignant hyperthermia.
- **Caution:** raised ICP, long QT, renal impairment.

KEY INTERACTIONS

- **Neuromuscular blockers:** enhanced effect.
- **Opioids, benzodiazepines:** reduce MAC requirement.
- **Adrenaline:** fewer arrhythmias than halothane.

NURSING MANAGEMENT

- PACU: monitor airway, SpO_2, RR, HR, BP and temperature.
- Watch for agitation in children; keep safe, reassure.
- Treat PONV; monitor for shivering.
- Recognise malignant hyperthermia: rising temp, rigidity, tachycardia — call for dantrolene.
- Make sure scavenging works in theatre.

PATIENT EDUCATION

- Children may be confused or upset on waking; this passes quickly.
- Mild nausea is common after anaesthesia.
- Tell staff about family history of anaesthetic reactions.

KEY POINTS

- Inhalational agent of choice for children (sweet, non-irritant).
- Rapid onset and recovery.
- Malignant hyperthermia trigger.
- Emergence delirium in children.

SOURCES CHECKED **[86]** Sevoflurane, inhaled, (inhalation) (US label)

Full references p. 432–457